Anna C. Nelson

CONTACT INFORMATION

Department of Mathematics & Statistics
The University of New Mexico
1 University of New Mexico, MSC01 1115
Albuquerque, NM 87131

Email: annanelson@unm.edu
Website: http://annacnelson.github.io
Office: SMLC 226

RESEARCH INTERESTS

Applied dynamical systems, mathematical biology, polymerization, mathematical modeling

ACADEMIC APPOINTMENTS

University of New Mexico, Department of Mathematics & Statistics — Albuquerque, NM
- Assistant Professor of Mathematics — January 2025 – present

Duke University, Department of Mathematics — Durham, NC
- Adjunct Assistant Professor — January 2025 – May 2025
- Phillip Griffiths Assistant Research Professor (postdoctoral) — August 2024 – December 2024
- William W. Elliott Assistant Research Professor (postdoctoral) — August 2021 – July 2024

EDUCATION

University of Utah, Salt Lake City, UT — May 2021
- Ph.D., Mathematics
- Thesis: Kinetic Polymerization Models and the Roles of Fibrinogen in Fibrin Gel Formation
- Advisor: Aaron Fogelson

Boise State University, Boise, ID — December 2012
- B.S., Applied Mathematics, *Summa Cum Laude*
- Minor: Computer Science

PUBLICATIONS

10. **A. C. Nelson**, E. F. Yao, Y. Zhang, C. V. Cook, S. Fischer-Holzhausen, L. Keeler Bruce, P. Dutta, S. Gholami, B. J. Smith, and A. N. Ford Versypt. "Mathematical modeling of bone remodeling after surgical menopause." *Frontiers in Systems Biology*, 6, 2026.
9. **A. C. Nelson**, S. A. McKinley, M. M. Rolls, and M. V. Ciocanel. "Emergent microtubule properties in a model of turnover and nucleation." *Journal of Theoretical Biology*, 616:112254, 2026.
8. H. G. Scanlon, G. Mahata, **A. C. Nelson**, S. A. McKinley, M. M. Rolls, and M. V. Ciocanel. "Nucleation feedback can drive establishment and maintenance of biased microtubule polarity in neurites."*Mathematical Biosciences*, 389:109538, 2025.
7. **A. C. Nelson**, M. M. Rolls, M. V. Ciocanel, and S. A. McKinley. "Minimal mechanisms of microtubule length regulation in living cells." *Bulletin of Mathematical Biology*, 86(58), 1-33, 2024.
6. **A. C. Nelson** and A. L. Fogelson. "Towards understanding the effect of fibrinogen interactions on fibrin gel structure." *Physical Review E*, 107(2):024415, 2023.
5. A. L. Fogelson, **A. C. Nelson**, C. Zapata-Allegro, and J. P. Keener. "Development of fibrin branch structure before and after gelation." *SIAM Journal on Applied Mathematics*, 82(1), 2022.
4. **A. C. Nelson**, M. A. Kelley, L. M. Haynes, and K. Leiderman. "Mathematical models of fibrin polymerization: past, present, and future." *Current Opinion in Biomedical Engineering*, 20 (100350), 2021.
3. **A. C. Nelson**, J. P. Keener, and A. L. Fogelson. "Kinetic model of two-monomer polymerization". *Physical Review E*, 101(2), 2020.
2. J. L. Herlin, **A. C. Nelson** and M. Scheepers. "Using ciliate operations to construct chromosome phylogenies". *Involve*, 9(1), 2016.

BOOK CHAPTERS

1. A. Kent, K. Leiderman, **A. C. Nelson**, S. S. Sindi, M. M. Stadt, L. Xiong, and Y. Zhang. "Studying the effects of oral contraceptives on coagulation using a mathematical modeling approach." In *Mathematical Modeling for Women's Health: Collaborative Workshop for Women in Mathematical Biology*, pages 83–132. Springer Nature, 2024.

PREPRINTS

J. Cruts, F. Gijsen, A. L. Fogelson, and **A. C. Nelson**. "Platelet plug microstructure and flow modulate fibrin gelation dynamics: Insights from computational simulations." *Submitted*, arXiv:2604.07844.

A. C. Nelson, Y. Zhang, M.V. Ciocanel, and C. Copos. "A multiscale model of size adaptation in actomyosin networks." *In preparation*.

AWARDS

Top 5% of Duke University undergraduate instructors, Trinity College — Fall 2023
For at least two of of the following categories: Overall quality of course, overall quality of instructor, intellectual stimulation of course.

Lewis Blake Award for Excellence in Teaching, Mathematics, Duke University — 2023
Annual postdoctoral award given for excellence in teaching.

BioFire Scholar Award, Mathematics, University of Utah — 2020
Annual award to one graduate student in department; includes stipend, tuition, and travel.

AWM Student Chapter Award for Scientific Excellence — 2020
One of four national awards given by the Association for Women in Mathematics while as Student Chapter Vice President.

FUNDING

Seed Grant, Duke Office for Faculty Advancement — February 2022 – March 2023
$14,000 award for Faculty-Student (FaSt) Math Series to build bridges and community among students and faculty. Grant aims include organizing events and programs such as book clubs, student professional development panels, faculty mentorship training, and invited speakers.

Travel Grants

NITMB Focused Research Group Award — 2027
Travel funding for collaboration at National Institute for Theory and Mathematics in Biology on "Incorporating and quantifying human behavior in epidemiological models"

AIM SQuaRE Grant — 2024, 2026, 2027
Travel funding for collaboration at Pasadena, CA on "Mathematical modeling and analysis to understand mechanisms of thrombosis and oral contraceptives" for three years

AMS MRC Collaboration Travel Grant — 2024
$800 to travel for Mathematical Research Community collaboration

AWM Travel Grant — 2023
$3500 to attend ICIAM 2023 in Tokyo, JP

Travel Awards

ASCB Travel Fund — 2025
$500 to attend Cell Bio 2025 in Philadelphia PA

Duke University Arts & Science Travel Fund — 2024
$1000 to attend JMM 2024 in San Francisco CA

AWM/NSF Travel Award — 2023
$1500 to attend AWM Research Symposium in Atlanta GA

SIAM Early Career Travel Award — 2023
$650 to attend SIAM Dynamical Systems 2023 in Portland, OR

MAA Project NExT Fellow — 2021 – 2023
$5000 to attend MAA Mathfest 2022 & 2023 and JMM 2023

SELECTED TALKS

Platelet plug microstructure and flow modulate fibrin gelation dynamics

SIAM Life Sciences, Invited Minisymposium — July 2026

Building community and finding connections in mathematics

Mathematics Colloquium, Cal Poly San Luis Obispo — May 2026
Bi-Co Math Colloquium, Bryn Mawr College and Haveford College — December 2025
Equity Forum, Montana State University — April 2025
Math For All Conference, Clemson, SC (**Plenary**) — April 2024

Mathematical models of polymerization processes in physiology

Mathematics Colloquium, Cal Poly San Luis Obispo — May 2026
Applied Mathematics Seminar, University of Notre Dame — April 2026
Applied Mathematics Seminar, Montana State University — April 2025
Biomath Seminar, Virginia Commonwealth University — March 2024

Mathematics Colloquium, University of Cincinnati	January 2024
Mathematical Biology Seminar, University of Illinois Urbana-Champaign*	December 2023
Biomath Seminar Series, NC State University	November 2023
Mathematical Biology Seminar, University of Pennsylvania	October 2023
Mathematical Biology Seminar, Brandeis University*	February 2023
Applied and Computational Mathematics Seminar, Tulane University	November 2022
Applied Math Seminar, Claremont Center for Mathematical Sciences*	October 2022

Modeling mechanisms of microtubule nucleation and polarity in living neurons

SMB/ECMTB, Invited Minisymposium	July 2026
NSF-Simons NITMB, Invited Workshop Presentation (20 minutes)	June 2026
AMS Spring Western Sectional Meeting, Invited Special Session	March 2026
Math Biology Seminar, Arizona State University	February 2026
Joint Mathematics Meeting, Invited Special Session	January 2026
ASCB Cell Bio 2025, Invited Minisymposium	December 2025
NSF-Simons NITMB, Invited Workshop Presentation (45 minutes)	November 2025
SIAM/CAIMS Annual Meeting, Invited Minisymposium	August 2025
AMS Spring Southeastern Sectional Meeting, Invited Special Session	March 2025
Joint Mathematics Meeting, Invited Special Session	January 2025

Modeling mechanisms of microtubule dynamics and polarity in living neurons

SIAM Annual Meeting, Invited Minisymposium	July 2024
Joint Mathematics Meeting, Invited Special Session	January 2024
10th ICIAM, Invited Minisymposium	August 2023
MAA MathFest, Invited Paper Session	August 2023
SMB Annual Meeting, Invited Minisymposium	July 2023
AMS Spring Central Sectional Meeting Invited Special Session	April 2023
Joint Mathematics Meeting, Invited AMS Special Session	January 2023

Towards a model of platelet aggregation and fibrin polymerization

Joint Mathematics Meeting, Invited AMS Special Session	January 2024
AWM Research Symposium, Invited Special Session	September 2023
AWM Research Symposium, Invited Special Session	June 2022

Kinetic polymerization models and the roles of fibrinogen in fibrin gel formation

Applied Mathematics Colloquium, University of North Carolina, Chapel Hill	April 2024
Mathematical Biology Seminar, University of California, Davis*	October 2021
Mathematical Biology Seminar, Duke University	September 2021
Mathematical Biology Seminar, U. of British Columbia & U. of Utah*	March 2021

Understanding the effects of fibrinogen interactions on fibrin gel structure

SIAM Conference on the Life Sciences, Invited Special Session	July 2022
SMB Annual Meeting, Invited Minisymposium*	June 2021
SIAM Conference on the Life Sciences, Invited Special Session*	June 2020

A kinetic model of two-monomer polymerization

Joint Mathematics Meeting, Invited AMS-AWM Special Session	January 2020
AMS Fall Western Sectional Meeting, Invited Special Session	November 2019
Boise State University Mathematics REU Program, Boise State University	July 2019

* Remote talk

INVITED WORKSHOPS

National Institute for Theoretical and Mathematical Biology, Chicago IL Extreme Events in Biological Functions	June 2026
National Institute for Theoretical and Mathematical Biology, Chicago IL Machine Learning of Cytoskeletal Machines (Cell Migration and Mitosis)	November 2025
ICERM, Brown University, Providence RI Patterns, Dynamics, and Data in Complex Systems	January 2025
National Institute for Theoretical and Mathematical Biology, Chicago IL Random Dynamical Systems with Applications in Biology	November 2024
AMS Mathematical Research Community, Java Center NY Complex Social Systems	June 2023

Banff International Research Station, Banff AB — March 2023
Sex Differences in Physiology: Mathematical Modelling and Analysis
Collaborative Workshop for Women in Mathematical Biology, Eden Prairie MN — June 2022
Mathematical Approaches to Support Women's Health,

MENTORSHIP

Graduate Research
Janneke Cruts, Erasmus Medical Center — Spring 2024 – Spring 2026
Hannah Scanlon, Duke University — Spring 2022 – Spring 2026

Undergraduate Research
Carson Dudley (undergraduate thesis), Duke University — Spring 2022 – Spring 2023
Maycol Vilchez, University of Utah (with Aaron Fogelson) — Spring 2020

Undergraduate Directed Reading Program, University of Utah — Spring 2019
Chase Stolworthy, use machine learning for predictions on voting data in Utah

AWM Undergraduate Mentor — 2019 – 2024
Paired with undergraduate students to meet monthly to discuss semester, future plans, and build community at University of Utah and Duke University.

SPIRE Fellows Postdoctoral Assistant and Faculty Mentor — 2021 – 2023
Assisted in organizing monthly events. and running academic support/mentoring system for high achieving undergraduates from historically excluded backgrounds. Taught course titled "Being Human in STEM at Duke", which is a discussion-based course on identity and humanity in STEM.

TEACHING EXPERIENCE

University of New Mexico
MATH 316, Applied Ordinary Differential Equations — Fall 2026
MATH 583, Methods of Applied Mathematics I† — Fall 2025

Duke University
MATH 353, Ordinary and Partial Differential Equations — Fall 2024, 2021
MATH 353, Ordinary and Partial Differential Equations — Spring 2024, 2022
MATH 270/BIO 218, Biological Clocks: How Organisms Keep Time — Fall 2023, 2022
MATH 577, Mathematical Modeling† — Spring 2023
MATH 75, Being Human in STEM for First Year SPIRE Fellows — Spring 2023, 2022

University of Utah
MATH 2250, Differential Equations and Linear Algebra♯ — Spring 2019
MATH 1030, Intro to Quantitative Reasoning‡ — Summer 2018
MATH 1220, Calculus II — Spring 2018
MATH 1100, Business Calculus — Fall 2017
MATH 1050, College Algebra — Summer 2017‡, Spring 2017, Fall 2016
MATH 1030, Intro to Quantitative Reasoning — Summer 2016‡, Spring 2016

† Graduate level course, ‡ Asynchronous online course, ♯ >100 students

Project NExT Fellowship — 2021 – 2023
Professional development program for early career mathematicians directed towards improving the teaching and learning of mathematics, fostering inclusivity in the mathematics community, and providing early career faculty strategies to engage in research, scholarship, and service opportunities.

Mathematics Instructor Training Facilitator, University of Utah — 2017, 2018, 2019
Facilitated annual workshop for new teaching assistants in the mathematics department. Responsibilities include organizing/planning workshops, observing new teachers, and giving lectures on teaching pedagogy.

SERVICE TO THE PROFESSION

Secretary, Society for Mathematical Biology — November 2024 – present
Elected position for Cell and Developmental Biology Subgroup
Journal referee
Mathematical Biosciences, Journal of Theoretical Biology, PLOS Computational Biology, SIAM Journal on Life Sciences
Conference session organizer
Minisymposium, ECMTB/SMB Annual Meeting, Graz AT — July 2026
"State of the art methods in modeling for cell and developmental biology"

Minisymposium, SIAM Life Sciences, Cleveland OH July 2026
"Mechanistic physiological modeling: A celebration of Aaron Fogelson's 70th Birthday"
Minisymposium, SIAM Life Sciences, Cleveland OH July 2026
"Computational and Mathematical Approaches to Cytoskeletal Dynamics"
Special Session, Spring AMS Western Sectional Meeting, Boise ID March 2026
"Modeling Complex Biological Systems on Multiple Scales"
Minisymposium, ASCB Cell Bio 2025, Philadelphia PA December 2025
"Quantitative Modeling Insights for Cytoskeleton Dynamics and Cell Polarity"
Minisymposium, SIAM Annual Meeting, Montréal QB August 2025
"Celebrating diversity in mathematical biology, with applications in medicine, physiology, and public health"
Special Session, SMB Annual Meeting, Edmonton AB July 2025
"From data to mechanisms: advancement in modeling in cell and developmental biology "
Special Session, Joint Mathematics Meeting, Seattle WA January 2025
"Diversity in Mathematical Biology"
Minisymposium, SIAM Annual Meeting, Spokane WA July 2024
"Modeling Dynamics in Biological Systems"
Minisymposium, AWM Research Symposium, Atlanta GA September 2023
"Promoting children's and women's health with mathematical and computational approaches"
Minisymposium, 10th ICIAM, Tokyo JP August 2023
"Recent Advances in Modeling Complex Systems and Multiscale Problems in Mathematical Biology"
Invited Paper Session, MAA MathFest, Tampa FL August 2023
"Recent Advances in Mathematical and Computational Biology, Highlighting Contributions from Undergraduate Researchers."
Minisymposium, SIAM Life Sciences, Pittsburgh PA July 2022
"Mathematical Modeling of Blood Clotting and its Application"
Minisymposium, SMB Annual Meeting, Virtual June 2021
"Mathematical Modeling of Blood Clotting: From Surface-Mediated Coagulation to Fibrin Polymerization"

Judge

SMB Poster Session, Edmonton AB July 2025
TriCAMS Poster Session, Chapel Hill NC October 2024
SIAM Annual AWM Graduate Student Poster Session, Spokane WA July 2024
JMM Undergraduate Student Poster Session, San Francisco CA January 2024
MAA MathFest Student Poster Session, Tampa FL August 2023
SIAM Dynamical Systems Red Sock Poster Session, Portland OR May 2023
MAA MathFest Student Poster Session, Philadelphia PA August 2022
JMM Undergraduate Student Poster Session, Denver CO January 2020

SERVICE TO THE UNIVERSITY

Organizer, UNM Applied Mathematics Seminar 2025 – present
Organization of biweekly research seminar for faculty, graduate students and postdocs.
Co-organizer, Duke Mathematical Biology Seminar 2022 – 2025
Organization of weekly research seminar for faculty, graduate students and postdocs.

Presenter

Grad-Fac Seminar, Department of Mathematics, Duke University October 2023
"The mathematics of bell-ringing"
Grad-Fac Seminar, Department of Mathematics, Duke University January 2023
"Mathematical modeling of polymerization processes in physiology"
SPIRE Speaker Series, Duke University August 2021
"Who can do math?"
Math Graduate Student Colloquium, University of Utah October 2020
"Computing in the Natural World: *In vivo* and *in vitro*"
Math Graduate Student Colloquium, University of Utah February 2020
"The mathematics of bell-ringing"

Organizer, Biofluids research seminar, University of Utah 2020 – 2021
Organization of weekly research seminar for faculty, graduate students and postdocs.

SERVICE TO THE COMMUNITY

Co-organizer, An Applied Mathematician's Storytelling Event
SIAM Annual Meeting, Cleveland OH — July 2026
SIAM Annual Meeting, Montréal QB — August 2025

Committee member, Mathematics DEI Team, Duke University — August 2022 – May 2024

Panelist, GROW (Graduate Research Opportunities for Women), Duke University — October 2022
"From day 1 to PhD"

Co-organizer, Duke Math Circles, Durham NC — August 2023 – January 2025
Manage volunteers and activities for Duke Math Circles program

Presenter, Girls Exploring Math, Duke University — June 2023
"Math: We R_0 afraid to use it!"

Volunteer, Duke Math Circles, Durham NC — August 2022 – April 2024
Provide exploratory instruction for K-6 students at Central Park School for Children

Panelist, Society for Women in Mathematics (SWiM), Colorado School of Mines — October 2020
"Graduate school panel" (virtual)

Co-organizer, Faculty-Student Weekly Tea, FaSt Grant — February 2022 – December 2023
Department of Mathematics, Duke University

Co-organizer, Faculty-Student Math Book Club — February 2022 – May 2023
Department of Mathematics, Duke University

Volunteer, Defining Your Path – Field Trip Program, University of Utah — February 2020

Judge, State of Utah Sterling Scholar Award, Mathematics, Salt Lake City UT — January 2020

Panelist, Clayton Middle School – Career Fair, Salt Lake City UT — January 2020

Co-chair, AWM Speaker series committee, Mathematics, University of Utah — 2020 – 2021
Invite and host mathematicians from underrepresented groups to give talks and network with department.

Vice President, AWM Student Chapter, University of Utah — 2019 – 2020
Organize monthly student events for undergraduates and graduate students, organize outreach events on and off campus, and meet with job candidates.

Presenter, Science Day at the U., University of Utah — November 2019
"Computing in Nature: Using DNA to solve math problems"

Presenter, Girls Full STEAM Ahead Camp, Leonardo Museum, Salt Lake City UT — July 2016
"Math: We R_0 afraid to use it!"

WORK EXPERIENCE

Bioinformatics Summer Intern — May 2019 – August 2019
Sera Prognostics, Salt Lake City, UT
Developed R scripts to remove batch and technical effects in proteomic data to aid in preterm birth prediction.

MEMBERSHIPS

American Mathematical Society
Association for Women in Mathematics
Society for Industrial and Applied Mathematics
Society of Mathematical Biology